

HDOC-12

Question:

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Find the sum and product of the first and last digits. **E.g.** Number: 12345, Sum = 6, Product = 5
- Count how many digits are lesser than, equal, and greater than a given digit 'k'. **E.g.** Number: 12345, k=3, Count_lesser=2, Count_equal=1, Count_greater=2

Task1: Algorithm:

Step 1: Start.

Step 2: Declare integer variables:

num, choice, first, last, sum, product, k, temp, digit, count_lesser, count_equal, count_greater.

Step 3: Input a positive integer num.

Step 4: Check whether num is positive.

- If num <= 0, display "Invalid input. Enter a positive integer." and stop.

Step 5: Display the menu:

1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k

Step 6: Input the user's choice.

Step 7: Use switch(choice) to perform the selected operation.

Case 1: Sum and Product of First and Last Digits

7.1: Find the last digit:

last = num % 10

7.2: Store the number in temp:

temp = num

7.3: Repeat while temp >= 10:

temp = temp / 10

7.4: The remaining digit is the first digit:

first = temp

7.5: Calculate:

sum = first + last

product = first * last

7.6: Display the first digit, last digit, sum, and product.

Case 2: Compare Digits with k

7.7: Input digit k.

7.8: Check whether k is between 0 and 9. If not, display an error message.

7.9: Initialize:

count_lesser = 0

count_equal = 0

count_greater = 0

temp = num

7.10: Repeat while temp > 0:

- Find the last digit: digit = temp % 10
- If digit < k, increment count_lesser.
- Else if digit == k, increment count_equal.
- Else, increment count_greater.
- Remove the last digit using temp = temp / 10.

7.11: Display count_lesser, count_equal, and count_greater.

Default Case

7.12: Display "Invalid choice."

Task2: test case table:

S.No.	Input Number	Choice	Extra Input (k)	Expected Output	Actual Output	Result
1	12345	1	-	Sum=6, Product=5	Sum=6, Product=5	Pass
2	12345	2	k=3	Less=2, Equal=1, Greater=2	Less=2, Equal=1, Greater=2	Pass
3	7	1	-	Sum=14, Product=49	Sum=14, Product=49	Pass
4	1000	2	k=0	Less=0, Equal=3, Greater=1	Less=0, Equal=3, Greater=1	Pass

Task3: C- program:

```
#include <stdio.h>
#include <math.h>
int main()
{
    long int num, temp;
    int choice;
    int firstDigit, lastDigit, sum, product;
    int k, digit, countLess, countEqual, countGreater;
    printf("Enter a positive integer: ");
    scanf("%ld", &num);
    /* Validate that the number entered is positive */
    if (num <= 0)
    {
        printf("Invalid input! Please enter a positive integer.\n");
        return 0;
    }
    printf("\n----- MENU ----- \n");
    printf("1. Sum and Product of first and last digit\n");
    printf("2. Count digits lesser, equal and greater than k\n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            temp = num;
            /* last digit = remainder when divided by 10 */
            lastDigit = num % 10;
            /* first digit = keep dividing by 10 till one digit is left */
            while (temp >= 10)
            {
                temp = temp / 10;
            }
            firstDigit = temp;
            sum = firstDigit + lastDigit;
            product = firstDigit * lastDigit;
            printf("\nNumber entered   : %ld\n", num);
            printf("First digit      : %d\n", firstDigit);
            printf("Last digit       : %d\n", lastDigit);
            printf("Sum of digits    : %d\n", sum);
            printf("Product of digits : %d\n", product);
            break;
        case 2:
            printf("Enter the digit k (0-9) to compare with: ");
            scanf("%d", &k);
            countLess = 0;
```

```

countEqual = 0;
countGreater = 0;
temp = num;
/* extract every digit one by one and compare with k */
while (temp > 0)
{
    digit = temp % 10;
    if (digit < k)
        countLess++;
    else if (digit == k)
        countEqual++;
    else
        countGreater++;
    temp = temp / 10;
}
printf("\nNumber entered : %d\n", num);
printf("Value of k    : %d\n", k);
printf("Count_lesser  : %d\n", countLess);
printf("Count_equal   : %d\n", countEqual);
printf("Count_greater : %d\n", countGreater);
break;
default:
    printf("\nInvalid choice! Please enter either 1 or 2.\n");
}
return 0;
}

```

Task4: output:

```
Enter a positive integer: 12345

----- MENU -----
1. Sum and Product of first and last digit
2. Count digits lesser, equal and greater than k
Enter your choice (1 or 2): 1

Number entered   : 12345
First digit      : 1
Last digit       : 5
Sum of digits    : 6
Product of digits : 5
```

```
Enter a positive integer: 12345

----- MENU -----
1. Sum and Product of first and last digit
2. Count digits lesser, equal and greater than k
Enter your choice (1 or 2): 2
Enter the digit k (0-9) to compare with: 3

Number entered : 12345
Value of k     : 3
Count lesser   : 2
Count equal    : 1
Count greater  : 2
```

```
Enter a positive integer: 7
Enter your choice (1 or 2): 1

Number entered : 7
First digit    : 7
Last digit     : 7
Sum of digits  : 14
Product of digits : 49
```

```
Enter a positive integer: 1000
Enter your choice (1 or 2): 2
Enter the digit k (0-9) to compare with: 0

Number entered : 1000
Value of k     : 0
Count lesser   : 0
Count equal    : 3
Count greater  : 1
```